Thermodynamic description of the Al–Bi system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Bi binary system as calculated with pycalphad from the SGTE dataset AlBi-06Kim.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L1 → (Al) + L2, lies at 658.6 °C.

1. Introduction

The Al–Bi system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of S.S. Kim, T.H. Sanders, Jr., Modelling Simul. Mater. Sci. Eng., 14, 1181-88(2006). The calculations use pycalphad 0.11.2 at 1 atm.

2. Phase diagram

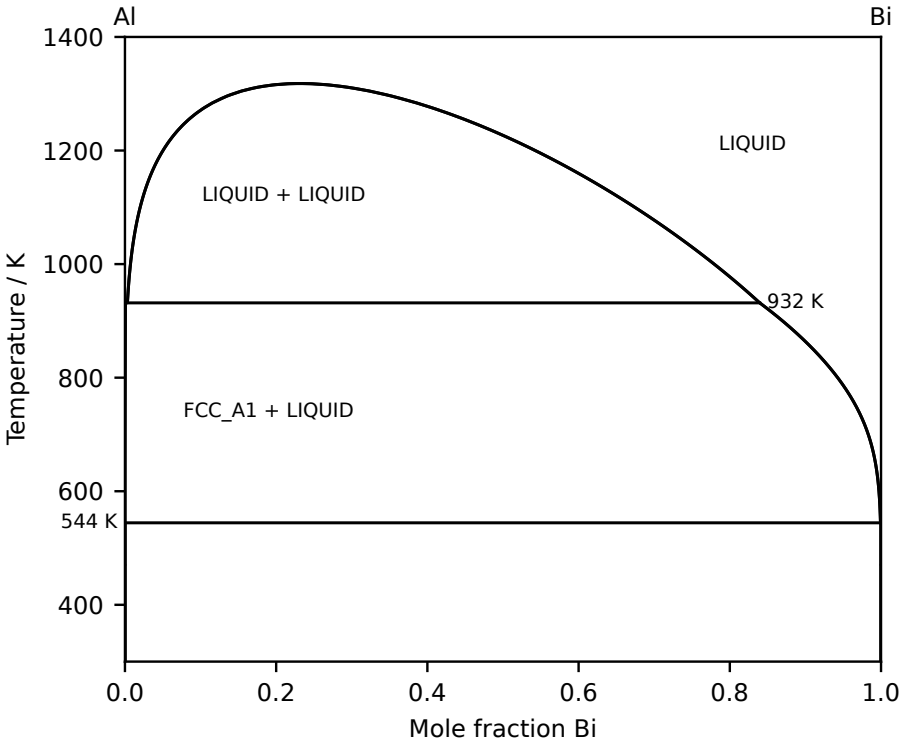


Fig. 1. Calculated Al–Bi phase diagram.

The dataset places a monotectic (L1 → (Al) + L2) at 658.6 °C; the composition of L1 is x(Bi) = 0.003. Eutectic equilibrium L → (Al) + (Bi) is found at 271.2 °C and a mole fraction of 0.999 (L).

Table 1. Invariant reactions in the Al–Bi system (calculated).

Reaction	Type	T / °C	Compositions (x(Bi))		
			phase 1	phase 2	phase 3
L1 → (Al) + L2	monotectic	658.6	0.003	0.001	0.840
L → (Al) + (Bi)	eutectic	271.2	0.999	0.000	1.000

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1400 K the integral mixing enthalpy of the liquid has its maximum, 19.17 kJ/mol, at x(Bi) = 0.440.

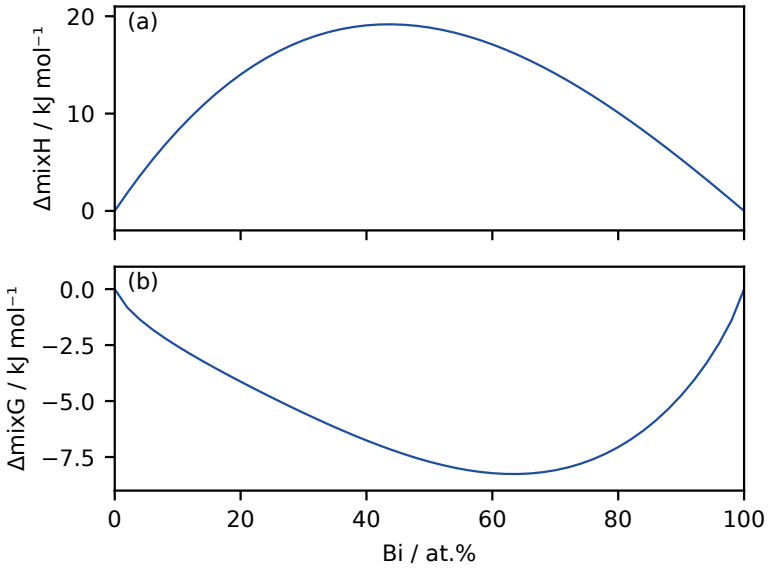


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1400 K.

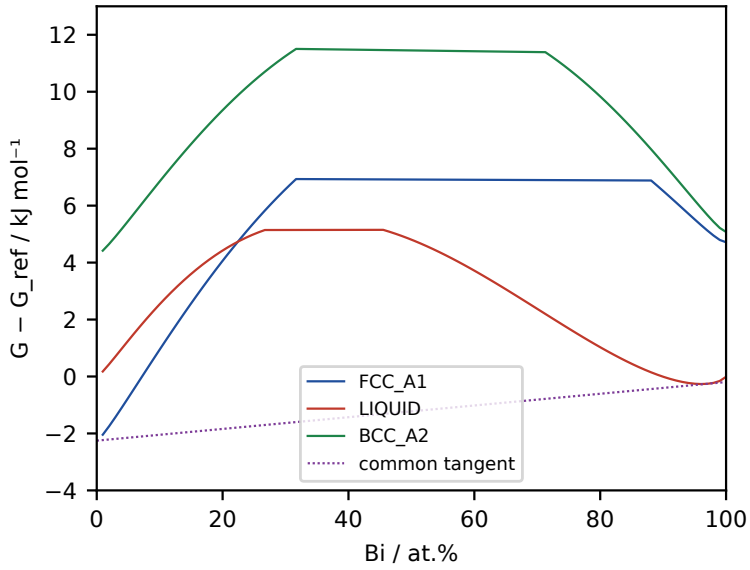


Fig. 3. Calculated Gibbs energy curves at 740 K.

Table 2. Redlich–Kister parameters of the liquid (Al–Bi).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	75351.7	-52.8	0.0
L1	20421.2	-3.7	0.0

Values as printed in the TDB file.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.